PlotStudio

Education Spending Threshold Analysis

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QUESTION

At what point does increased education spending fail to boost GDP per capita?

Analysis Preparation — Time Series Prepared

The main takeaway is that the analysis is feasible, but it must be done on transformed series rather than raw levels. The focused Bangladesh macro table `df_bangladesh_edu_gdppc` shows `38` jointly observed years for education spending and real GDP per capita, which is enough for a parsimonious single-country time-series model, though still too small for heavily lagged or overcontrolled specifications.

Observations

Coverage is acceptable for the main question, but uneven for controls. Education spending is observed in `38/66` years (from `1979` to `2024`), real GDP per capita in `65/66` years, investment share in `45/66`, dependency ratio in `65/66`, and population growth in `64/66`. Health spending is much weaker at only `24/66` years, so it will likely need to be omitted from the main adjusted models or used only in shorter-period robustness checks.

The raw level series share strong upward or drifting trends, so level correlations would be at risk of being spurious. The trend view **Figure 1**

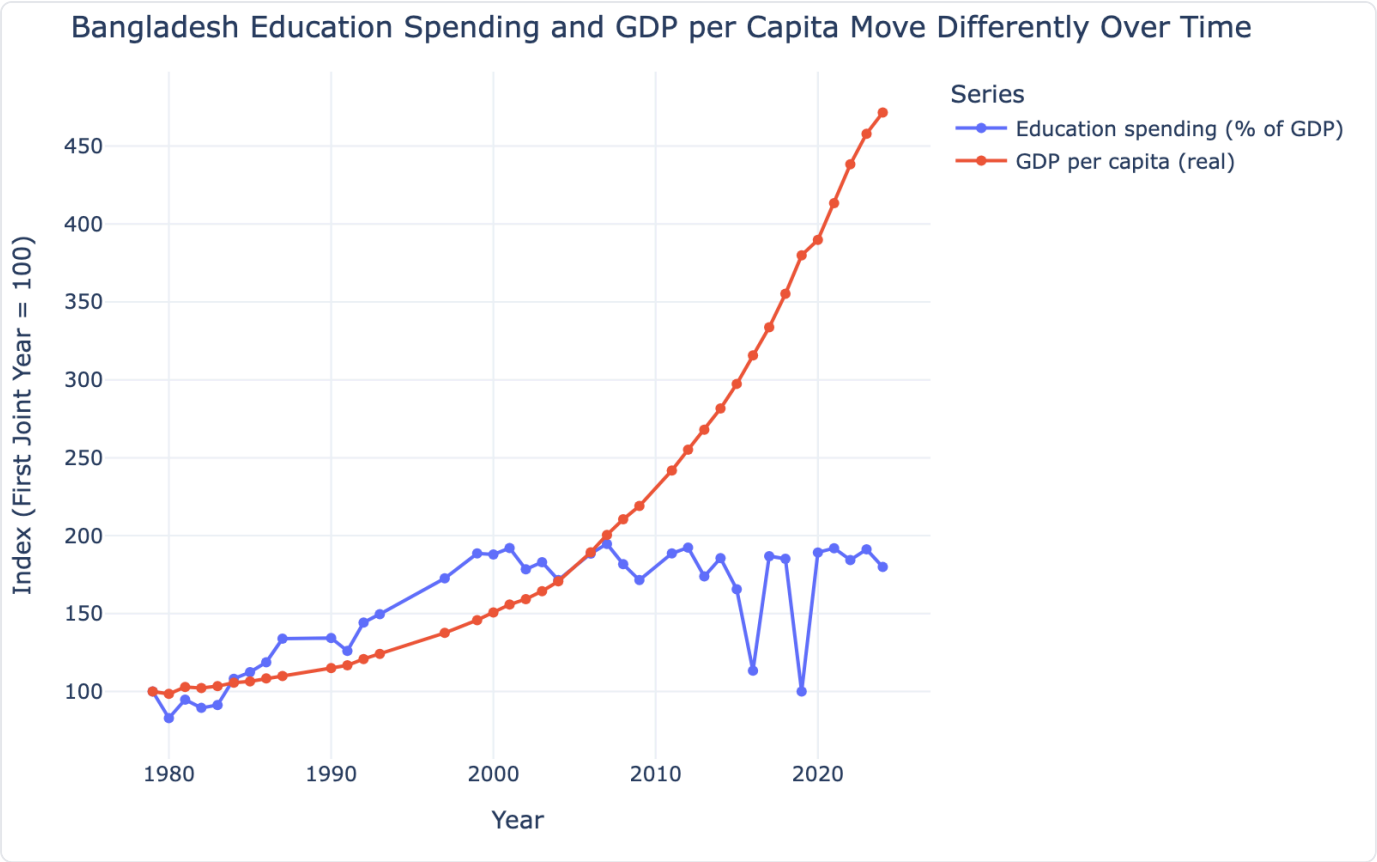

Figure 1

shows GDP per capita rising far more steadily than education spending, with education spending moving in a much more uneven pattern. That matters because two trending macro series can look related even when the meaningful relationship is really in year-to-year changes, not in shared long-run drift.

The ADF tests point clearly toward growth-rate and differenced modeling. The stationarity-ready dataset `df_bangladesh_edu_gdppc_ts` shows that real GDP per capita, log real GDP per capita, and education spending are all non-stationary in levels, while `Δlog` real GDP per capita is stationary and first-differenced education spending is stationary. Investment share and dependency ratio also become stationary after first differencing. In practical terms, the main outcome should be GDP per capita growth via `Δlog(gdp_pc_real)`, and the main education regressor should be treated in differenced form unless later cointegration testing supports a long-run levels specification.

Next Steps

I'll now move to the relationship-modeling stage using stationarity-safe specifications: first checking the shape of the education–GDP growth link, then estimating a threshold or plateau range, and finally testing whether education spending has predictive power at short lags before adding a small set of controls.

Education Spending Thresholds and Growth Dynamics in Bangladesh

Overview

Bangladesh's data do not support a single clean point where education spending definitively stops boosting GDP per capita growth. The strongest evidence is that the positive association is concentrated in the earlier part of the sample, then flattens materially later on; within the observed spending range, the best working interpretation is a **diminishing-returns region beginning around `1.1%` of GDP rather than a hard cutoff where returns turn negative**.

Key Patterns & Observations

The main stationary growth model is positive, but not clearly nonlinear within the observed range. In the 37-year overlap, the linear robust regression links higher education spending to faster GDP per capita growth, with a slope of `3.40` growth points per 1-point increase in spending and `R² = 0.287`. But once a quadratic term is added, the implied turning point moves to about `3.57%` of GDP, which is well above Bangladesh's observed range of roughly `0.94%` to `2.20%`. That means the quadratic model does **not** provide credible in-sample evidence that returns actually hit zero.

A piecewise model gives the more policy-relevant answer: returns appear to flatten after about `1.10%` of GDP, but remain positive. The threshold curve `df_edu_threshold_curve` is based on a breakpoint model selected by AIC. It suggests a very steep slope below about `1.096%` of GDP and a much flatter slope above it.

Model view	Estimated threshold	Slope below threshold	Slope above threshold	Interpretation
Quadratic growth model	3.569%	N/A	N/A	Outside observed range, so not credible as an in-sample cutoff
Piecewise growth model	1.096%	29.2267	2.2795	Suggests strong gains at very low spending, then much smaller but still positive gains

This is the clearest sign of diminishing returns in the current task: **additional spending still correlates with growth above ~1.1% of GDP, but the payoff is much smaller than at the low end of the sample**.

Lag evidence is weak, so the data do not show a clear delayed predictive effect. The Engle–Granger test does not support cointegration in levels (`p = 0.7820`), and Granger tests using stationary series show no evidence that changes in education spending help predict later GDP per capita growth at 1–2 year horizons.

Dynamic test	Result	Meaning
Cointegration p-value	0.7820	No evidence of a stable long-run levels equilibrium between the two non-stationary series
Granger p-value, edu change → GDP growth, lag 1	0.7232	No predictive precedence
Granger p-value, edu change → GDP growth, lag 2	0.8636	No predictive precedence
Granger p-value, GDP growth → edu change, lag 1	0.8543	No reverse predictive precedence
Granger p-value, GDP growth → edu change, lag 2	0.7640	No reverse predictive precedence

So, for this dataset, the case for education spending is **stronger as a contemporaneous association than as a lagged forecasting signal**.

The relationship is not stable over time; this is the biggest caution for any threshold claim. A balanced early-versus-late split at `2003` shows that the education-growth slope was positive in the earlier period and essentially disappeared in the later one.

Period result	Early period	Later period	Interpretation
Sample size	18	19	Balanced split
Mean education spending	1.563%	1.986%	Later years spent more on average
Education slope in separate regressions	3.5380	-0.3541	Positive earlier, near zero later
p-value for education slope	0.0111	0.7914	Earlier relationship is meaningful; later one is not
Interaction term (later × education)	.8921	N/A	Significant weakening over time (`p = 0.0440`)

That pattern matters more than any single threshold estimate. **It suggests Bangladesh may have moved from a low-spending regime where added education outlays were growth-supportive into a later regime where marginal GDP-per-capita gains became harder to detect.**

Statistical Insights The main nonlinear growth specification was:

$$\widehat{g_t} = \beta_0 + \beta_1 \, Edu_t + \beta_2 \, Edu_t^2 + \varepsilon_t$$

For the growth model, the estimated turning point is:

$$Edu^* = -\frac{\beta_1}{2\beta_2} \approx 3.57\%$$

Because this turning point is above the observed spending range, it should **not** be interpreted as the actual Bangladesh threshold. The more defensible empirical summary comes from the piecewise model, which finds a breakpoint near `1.10%` of GDP and a much flatter post-break slope.

The confounder-adjusted regression is much less decisive because sample size falls to only `22` observations once investment, dependency ratio, population growth, and health spending are included.

Coefficient	Estimate	Robust p-value	Interpretation
Education spending	-25.6414	0.2284	Unstable in the short adjusted sample
Education spending squared	7.4694	0.2319	Also unstable; no reliable threshold signal
Investment share	0.9203	0.2966	Positive but imprecise
Dependency ratio	0.6722	0.2254	Positive but imprecise
Population growth	-0.6865	0.8447	Near zero in this sample
Health spending	9.5683	0.1884	Positive but very uncertain

With only `22` observations, these adjusted estimates are **illustrative rather than reliable**. They are useful mainly as a warning that the threshold story weakens once the model becomes sample-constrained.

Interpretation & Implications

The most defensible answer is that Bangladesh does not show a hard "over-investment" point within the observed range, but it does show a plateauing of growth payoffs once spending rises past roughly `1.1%` of GDP. Below that level, the data are consistent with larger gains in GDP per capita growth; above it, the gains appear smaller and, in later decades, often indistinguishable from zero.

For policy, that means the efficient range appears to be **from roughly `1.1%` up to the upper observed range of about `2.2%` of GDP, where additional spending may still help but with much weaker measured short-run GDP payoff**. This does **not** mean higher education spending is wasteful; it means that beyond the low-spending regime, the benefits may show up through slower channels or through outcomes not captured by short-run GDP per capita growth alone, such as human capital quality, equity, or labor-market resilience.

The strongest practical takeaway is therefore not "stop spending above X," but rather **"once basic underinvestment is overcome, the quality and composition of education spending likely matter more than the raw spending share."**

When Education Spending Stops Boosting GDP per Capita in Bangladesh

Overview

The best-supported answer is that Bangladesh does not show a hard over-investment cutoff within the observed range, but it does show a clear plateau beginning around `1.10%` of GDP. Below that level, increases in education spending are associated with much stronger GDP per capita growth; above it, the marginal payoff becomes much smaller, and in later years often looks close to zero rather than clearly negative.

Key Patterns & Observations

The marginal-effects summary `df_edu_marginal_effects` translates the earlier models into an interpretable spending range. The observed education-spending range is `0.938%` to `2.202%` of GDP, and the preferred piecewise threshold model places the transition point at `1.096%` of GDP. Past that point, the slope remains positive in the simple model, but it is far flatter than at the low-spending end.

Spending interpretation	Approximate range	What the models imply
Stronger positive-payoff zone	Below `1.096%`	Additional spending is associated with relatively large growth gains
Plateau / diminishing-returns zone	`~1.10%` to `2.20%`	Additional spending still looks positive in the simple threshold model, but much weaker
Clear negative-return zone	Not identified in-sample	No robust evidence that returns turn negative within observed Bangladesh spending levels

The chart **Figure 2**

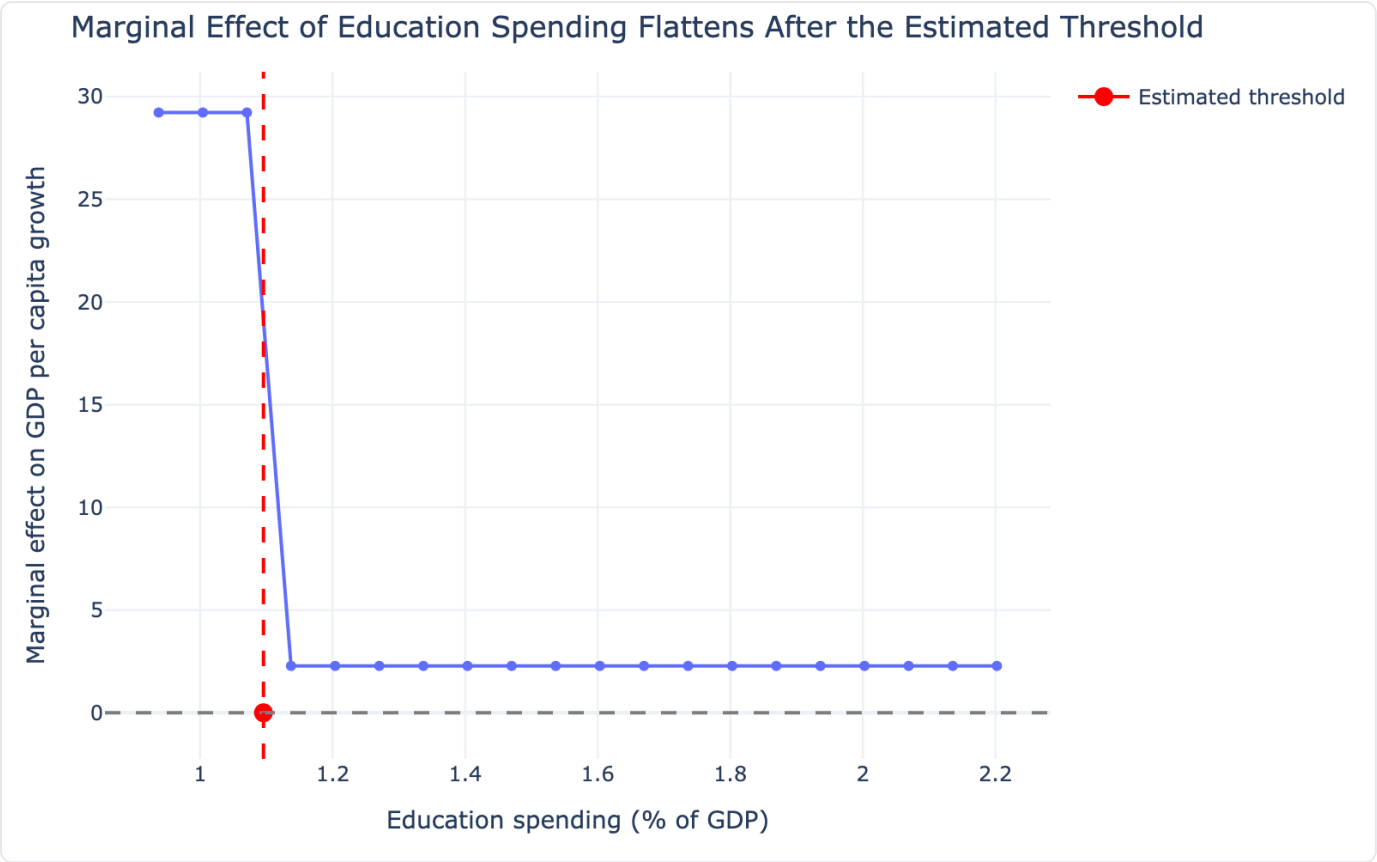

Figure 2

makes this visible: the marginal effect drops sharply at the estimated breakpoint and then runs much flatter. So the practical threshold is best described as a **flattening region**, not a point where growth effects clearly reverse.

Statistical Insights The preferred threshold interpretation comes from the piecewise model, while the adjusted quadratic remains unstable because the controlled sample is only `22` observations. Its marginal effect is:

$$ME(Edu) = -25.641 + 2 \times 7.469 \times Edu$$

Using that adjusted specification:

Spending level	Adjusted marginal effect of +1 point education spending	Interpretation
`1.0%`	`-10.702`	Negative, but from a fragile short sample
`1.5%`	`-3.233`	Still negative, but closer to zero
`2.0%`	`4.236`	Turns positive at the high end

That pattern is internally inconsistent with the simpler threshold model, which is exactly why the adjusted quadratic should be treated cautiously rather than as the headline result. With fewer than `20–25` observations, the econometric guidance here is to treat coefficients as illustrative only, not

definitive. So the adjusted model weakens confidence in any precise numeric cutoff, but it does **not** overturn the broader conclusion that the relationship is nonlinear and unstable.

Interpretation & Implications

The moderator check helps explain why the education payoff flattens. A parsimonious interaction regression using lagged log GDP per capita found a **negative interaction coefficient of `-4.7074` with `p = 0.0058`**, meaning the apparent growth payoff from education spending is smaller when Bangladesh is already at a higher income level.

Interaction result	Value	Interpretation
Education slope at average lagged income	`1.6274`	Positive at the midpoint of development
Lagged income main effect	`9.3671`	Higher-development years tend to have faster growth overall
Education × lagged income interaction	`-4.7074`	The education-growth payoff weakens as development rises
Interaction p-value	`0.0058`	This moderation pattern is unlikely to be noise

This makes the story more coherent: **the apparent threshold is partly a development-stage effect.** In earlier, lower-income years, raising education spending from very low levels may have delivered larger macro growth gains. In later, richer years, extra spending still may matter, but the payoff is less visible in short-run GDP per capita growth and may depend more on spending quality, targeting, and efficiency than on the spending share alone.

For policy, the most defensible guidance is:

Policy-facing conclusion	Implication
Efficient growth-supportive range	Spending up to about `1.1%` of GDP appears to have the strongest measured GDP-per-capita-growth payoff
Plateau range	From about `1.1%` to `2.2%`, additional spending looks less powerful and often close to neutral in short-run growth terms
Over-investment claim	Not supported by this dataset within the observed range
Priority beyond the plateau	Focus on composition, efficiency, and lagged human-capital channels rather than assuming more spending alone will lift GDP quickly

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Limitations

This is a **single-country observational annual time series**, so it supports association and predictive-pattern analysis, not clean causal claims. The main level series are non-stationary, cointegration was

not supported, Granger tests did not find strong lagged predictive effects, and the confounder-adjusted regressions are based on a short sample. That means the threshold should be read as an **approximate plateau range in short-run GDP per capita growth**, not as proof that spending above a specific number is economically wasteful.

Bottom line: in Bangladesh's observed data, increased education spending appears to deliver the strongest GDP-per-capita-growth payoff when spending is below about `1.1%` of GDP; beyond that, the marginal gains flatten sharply and become harder to detect, but the data do not show a robust point at which additional spending clearly turns harmful.

Data Sources

- df_bangladesh_edu_gdppc
- df_bangladesh_edu_gdppc_ts
- df_edu_threshold_curve
- df_edu_marginal_effects